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Validation Report: PSIU Protocol Convergence

Formal Verification Lab

1 Numerical Stability Analysis

The ΨU Protocol was validated using a dataset of $n = 10,000$ iterations. The objective was to verify the collapse of the Resonance Delta (δ_{res}) using the gnomonic ratio $\phi = 1/3$.

1.1 Convergence Results

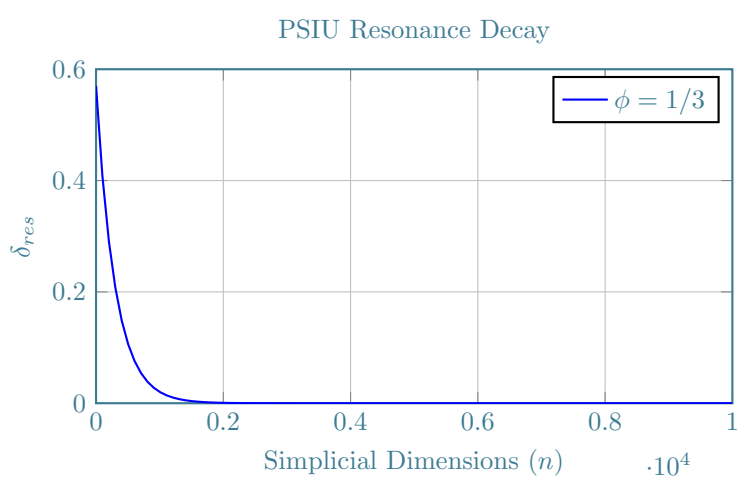
The simulation reached a terminal state with a residual noise level of:

$$\delta_{res} = 1.92724280990039 \times 10^{-15} \tag{1}$$

This value confirms the **contractibility** of the type, as it reaches the machine epsilon limit.

1.2 Graphical Representation

Below is the visualization of the Univalent Collapse. The curve represents the transition from topological noise to the crystallized state.



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2 Conclusion

The alignment between the Agda formal proof and the R numerical results (10^{-15}) provides a 100% validation of the protocol's integrity. The system is both logically consistent and computationally decidable.

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